

SymbioLink AI

AI-Powered Industrial Symbiosis & Circular-Economy Platform

Innovation Summary

● Problem Statement

Industrial clusters generate byproducts that could serve as another factory's raw material, but no digital system exists to discover such matches. A 2010 study of the Nanjangud industrial cluster found 99.5% on-site waste recovery, yet only 11 verified inter-firm exchanges across 60+ facilities: usable material is landfilled or down-cycled while nearby MSMEs pay full price for the same material as virgin input.

● Proposed Solution

SymbioLink AI models an industrial cluster as a graph, finding direct matches and 2-3 hop exchange chains ranked by saving, distance, and trust. Listings are created through a web form, a WhatsApp message in plain or regional-language text, or a simulated IoT bin sensor. The platform runs the full transaction — order lifecycle, seller confirmation, logistics pooling, payment, and two-way rating — plus carbon-credit issuance and BRSR-ready ESG reporting.

● Use of Artificial Intelligence

Five trained scikit-learn models: an **intake classifier** (95.6% held-out accuracy) parses free-text/WhatsApp listings; a **match-ranker** (71.0% accuracy / 0.725 AUC) scores every candidate exchange by saving, distance, and trust; a **forecasting model** (+14.2% MAE over baseline) predicts a unit's future waste volume; an **order-risk model** flags ring-trading patterns; a **photo classifier** grades material type/quality from an uploaded image. **Google OR-Tools (CP-SAT)** jointly optimizes the best non-overlapping match set across the whole cluster, rather than ranking chains independently.

● Innovation / Novelty

Multi-hop symbiosis matching rather than direct pairs only; a verified trust ledger built from real completed-exchange ratings; multilingual WhatsApp/text intake removing app-adoption and literacy barriers; simulated IoT-sensor auto-listing; logistics pooling across multiple orders on a route; and an integrated carbon-credit marketplace linked to live BRSR ESG reporting — combined in one platform rather than scattered across separate tools.

● Target Users / Beneficiaries

MSME and mid-size factory owners in industrial clusters (metal, textile, paper, chemical, food-processing, and similar sectors); industrial-cluster/estate associations; sustainability and compliance officers; and, indirectly, regulators and ESG-focused investors.

● Expected Impact

In a seeded 27-unit cluster simulation: roughly **Rs 22,845** saved and **1,104 kg** of CO₂e avoided across just 10 matches, figures that scale with cluster size. Lower input costs for MSMEs, wider participation in the formal circular economy through low-barrier intake, and audit-ready BRSR emissions reporting at cluster scale.

● Current Development Stage

Working prototype: Flask backend on PostgreSQL, deployed live on Render, with 5 trained ML models across 18 integrated modules (matching, trust, logistics pooling, IoT, carbon credits, BRSR reporting, geofencing, OR-Tools optimization, and more), backed by an automated pytest suite. **Pre-pilot** — not yet tested with real external companies.